

Bangladesh University of Engineering and Technology

CSE208: Data Structure and Algorithms Sessional II

Online - 1 on Minimum Spanning Tree

Section: B

Time: 30 Minutes

Village Power Grid

A local government wants to bring electricity to **N remote villages** in a hilly region. Power cables can be laid between certain pairs of villages, each at a known cost. Your task is to connect **all villages** so that electricity can reach every village — directly or indirectly — at **minimum total cost**.

Input Format

- First line: Two integers N (villages) and M (possible cable routes).
- Next M lines: Three integers $u \ v \ w$ — a cable between village u and v at cost w .

Output Format

- Print the minimum total cost, then each selected route as $u \ v \ w$ (one per line).
- If it is impossible to connect all villages, print `Not possible`.

Constraints

- $2 \leq N \leq 100,000$
- $1 \leq M \leq 500,000$
- $1 \leq w \leq 1000$ | Undirected, no self-loops.

Requirement

- No STL graph libraries. You may use `sort()`, `vector`, and basic STL only.

Samples

Sample 1

Input	Output
6 9 1 2 4 1 3 3 2 3 2 2 4 5 3 4 6 3 5 7 4 5 3 4 6 8 5 6 4	Minimum cost: 17 2 3 2 1 3 3 4 5 3 5 6 4 2 4 5

Sample 2

Input	Output
3 1 1 2 5	Not possible

Explanation: With only one cable, villages 1 and 2 are connected but village 3 is unreachable.